



TITLE PAGE

- Problem Statement ID- SIH1592
- Problem Statement Title- Student Innovation
- Theme- Clean & Green Technology
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)- AKY



IDEA TITLE

IDEA/SOLUTION: greenSort AI – “A Cleaner future starts with us”

greenSort AI is a web-based application that will serve as a platform that connects community, local authorities, and businesses in a unified effort to maintain cleanliness and environmental sustainability. The prototype will function as follows:

- Identify the bio-degradable wastes through image screening using **Artificial Intelligence (AI)** .
- It utilizes a **Convolutional Neural Network (CNN)** architecture enhanced by transfer learning from 2000+ pre-trained images which is used to detect garbages and plastics..
- Users generate **revenue** in the form of **cashback** which can be used in forms like discounts,coupons, prizes from local business as well as our partner businesses. It can also be used in transactions of **cryptocurrency and NFTs**.
- The nearest existing trader to the customer's location is assigned to collect the wastes through **path optimization**.
- People can also **purchase recycled products and other DIY items** manufactured from reusable plastic from our website at **cheaper prices**.

Problem Resolution:

- greenSort delivers a comprehensive platform that empowers citizens to report waste-affected areas, utilizes **IoT and AI** for optimizing waste collection and segregation through **smart technologies**.

Unique Value Propositions:

- **Real time monitoring and optimization of wastes by Municipal Corporation.**
- **Reduces labor resulting enhanced efficiency and cost-effectiveness.**
- **Multilingual Support.**

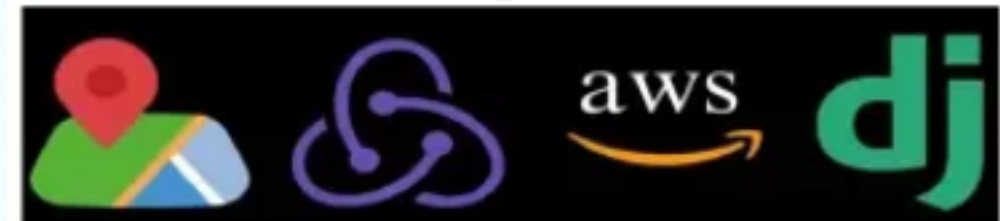
Web Application Tools:



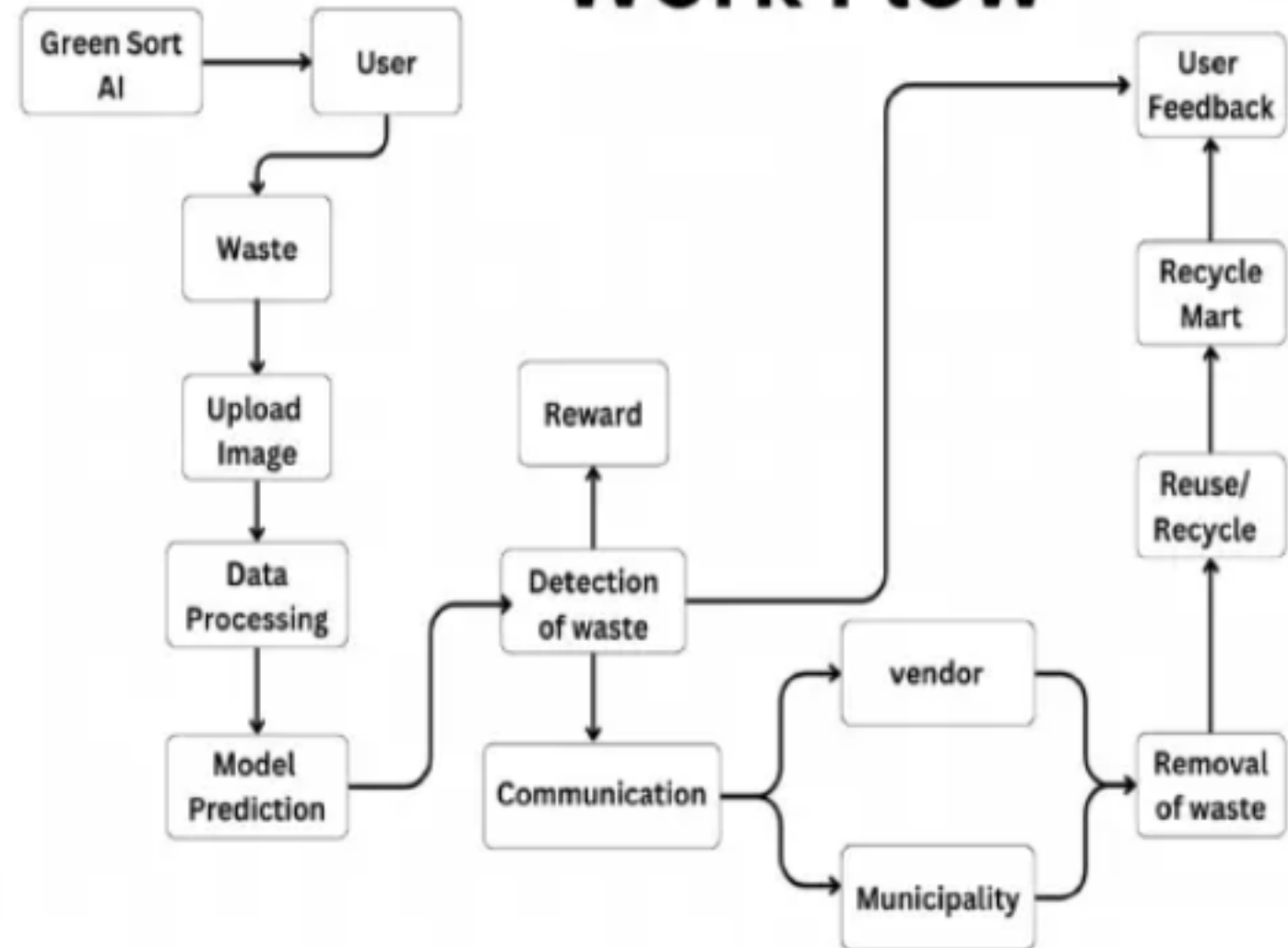
Artificial Intelligence and Machine Learning:



Additional Technologies:



Work Flow



Feasibility of the Idea:

- **Technical Feasibility:** Utilizes reliable technologies (**React, Node.js, TensorFlow**) for effective implementation and **AI-driven waste management**.
- **Operational Feasibility:** Successful deployment with adequate infrastructure and **user-friendly design**, supported by **stakeholder engagement** through incentives and training.
- **Economic Feasibility:** Initial costs are offset by **long-term savings, e-commerce feature**, charging a **minimal fee from vendors**.

Potential Challenges, Risks and its overcomings:

- **Technical Challenges:** Integration of various technologies and maintaining AI model accuracy could be problematic.
- ❖ **Solution:** Implement **comprehensive testing** on over 2000+ images and leverage scalable infrastructure to handle integration and accuracy issues.
- **Operational Challenges:** User adoption may face resistance due to change or unfamiliarity, and managing data security is crucial.
- ❖ **Solution:** Offer extensive user training and robust **data security** measures to facilitate adoption and protect data.
- **Economic Risks:** There is a risk of budget overruns and uncertain ROI affecting financial viability.
- ❖ **Solution:** Develop detailed budgets with contingency plans and monitor ROI to ensure financial sustainability.

IMPACT AND BENEFITS



SMART INDIA
HACKATHON
2024

Enhanced Efficiency:

- Impact: Optimized routes and automated categorization
- Benefit: Reduces costs, minimizes waste, boosts recycling.

Improved Sanitation:

- Impact: Predictive maintenance and real-time monitoring.
- Benefit: Cleaner streets and healthier communities.

Increased Engagement:

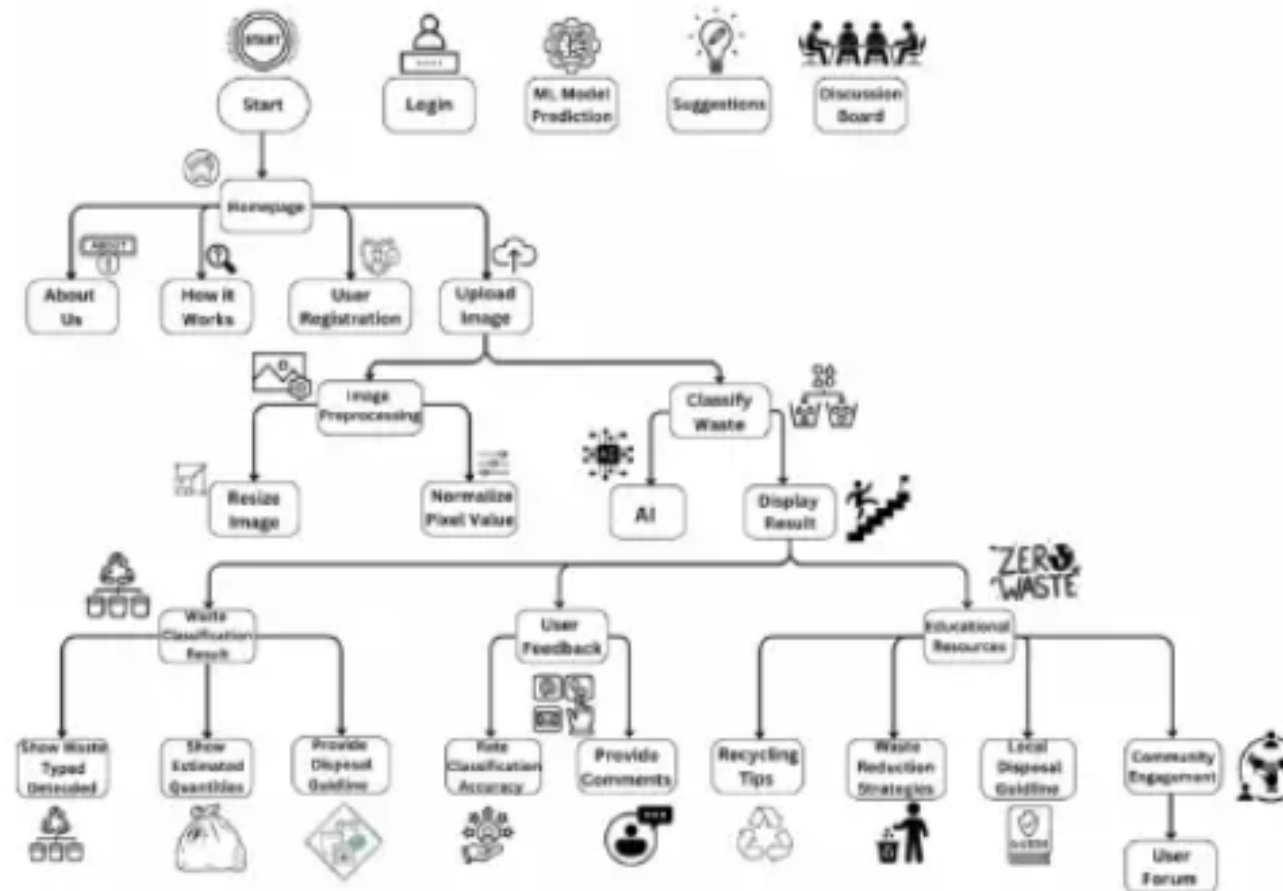
- Impact: Residents report issues, track reports, earn rewards.
- Benefit: Higher community involvement and responsibility.

Data-Driven Decisions:

- Impact: Comprehensive data and analytics.
- Benefit: Better management and strategic planning.

Scalability and Integration:

- Impact: Flexible architecture and external service integration.
- Benefit: Adapts to growth and technological advancements.



- Data research:

- <https://ourworldindata.org/waste-management>
- <https://www.sciencegate.app/keyword/229345>
- <https://developers.google.com/maps>

- Technology research:

- <https://react.dev/>
- <https://pytorch.org/>
- <https://www.tensorflow.org/>
- <https://opencv.org/>

